# Exploring a connection between air quality and labour productivity

Andrej Ohrablo | Světová ekonomika a podnikání II. 24/25

# Air quality




**Hypothesis:**

Air quality and labour productivity are correlated on global scale

# Air quality index

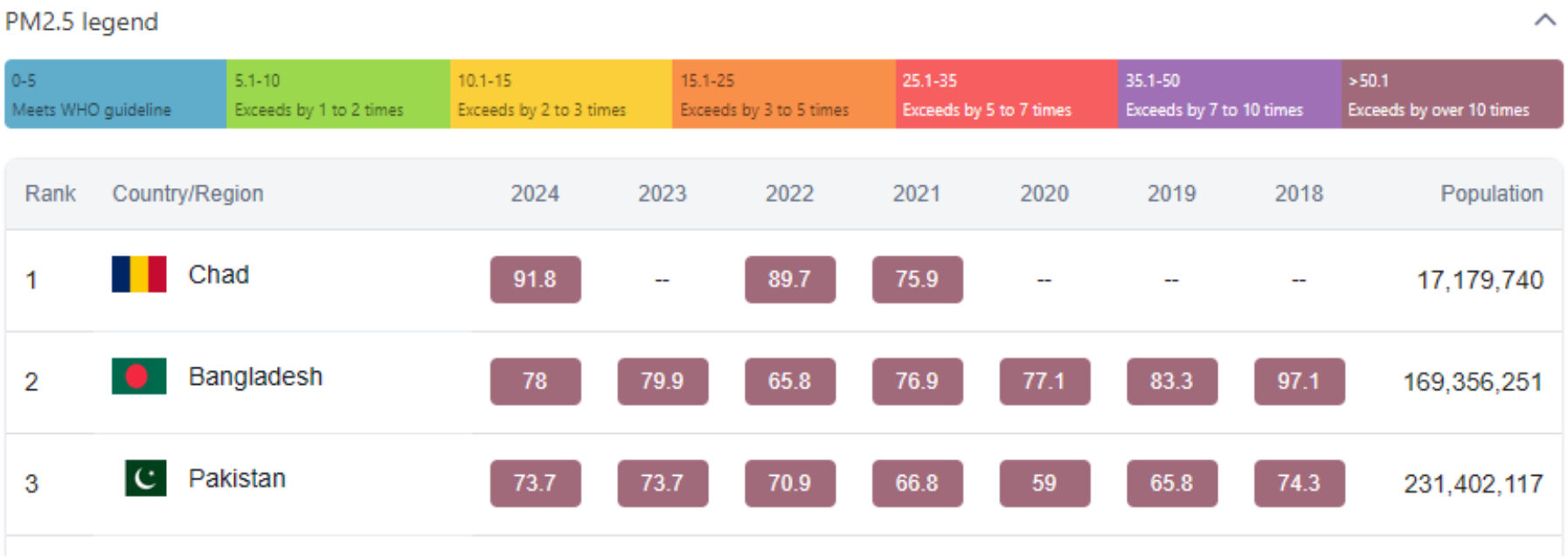
First index 1966





# Air quality index

- Measuring pollutant concentrations
- SO2, NO2, CO, O3, **PM2.5**, PM10 …
- Many different indexes, **IQAir**, EAQI




| Rank | Country/Region | 2024 | 2023 | 2022 | 2021 | 2020 | 2019 | 2018 | Population |
|---|---|---|---|---|---|---|---|---|---|
| 1 | Chad | 91.8 | -- | 89.7 | 75.9 | -- | -- | -- | 17,179,740 |
| 2 | Bangladesh | 78 | 79.9 | 65.8 | 76.9 | 77.1 | 83.3 | 97.1 | 169,356,251 |
| 3 | Pakistan | 73.7 | 73.7 | 70.9 | 66.8 | 59 | 65.8 | 74.3 | 231,402,117 |

# Labour productivity

- Labour productivity – GDP/hour worked
- Labour productivity → Air Quality : public policies, industry, low infrastructure
- Air Quality → Labour productivity : health benefits, less sick days, better cognitive skills

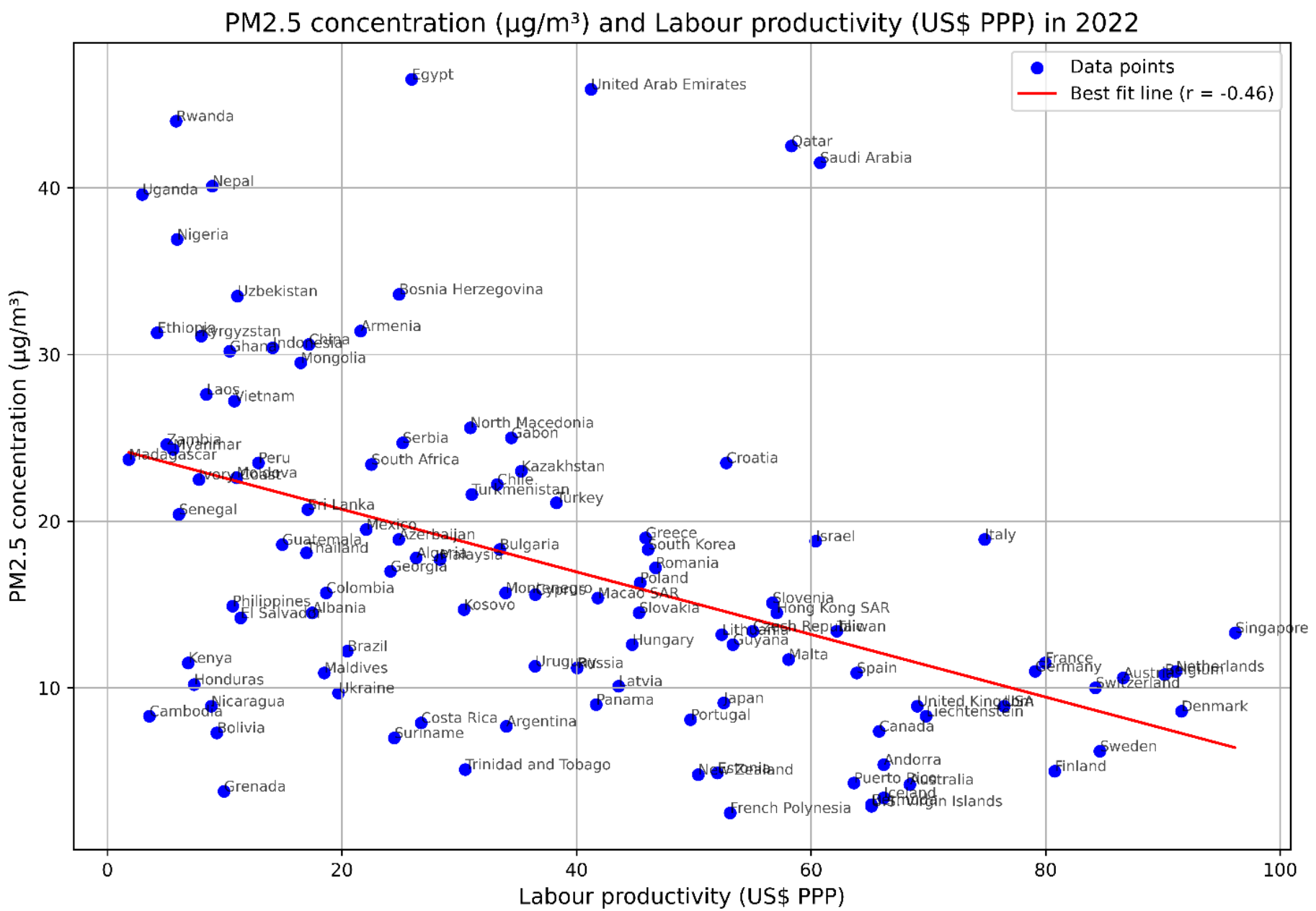
# Previous studies

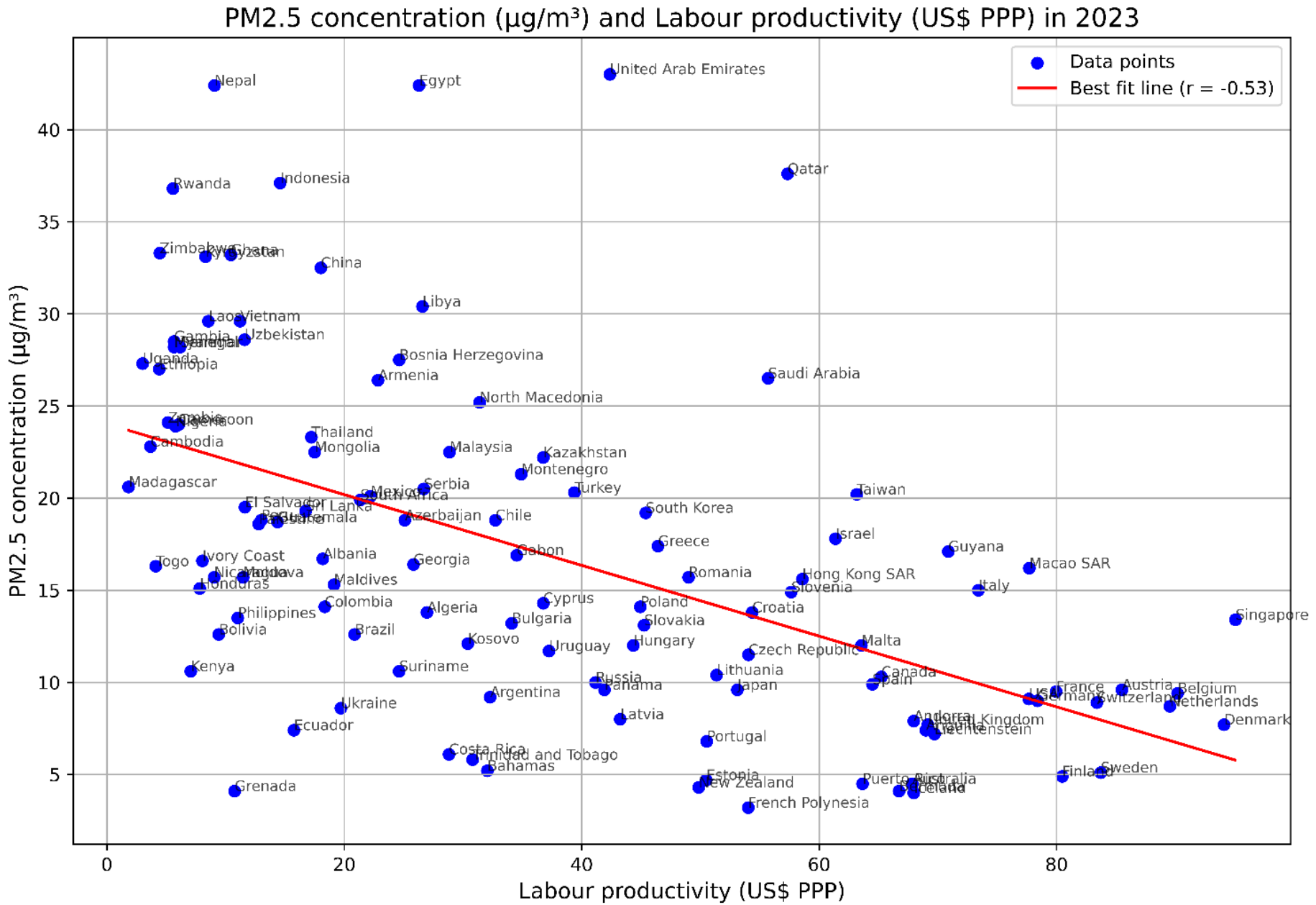
- **EU** - 1µg/m3 increase in PM2.5 causes a 0.8% reduction in real GDP
- **China** - a 10-unit increase in the air pollution index decreases the number of daily calls handled by a worker by 0.35 percent
- **France** – a 10% increase in fine particulate matter leads, on average, to a 1.5% decrease in labour productivity

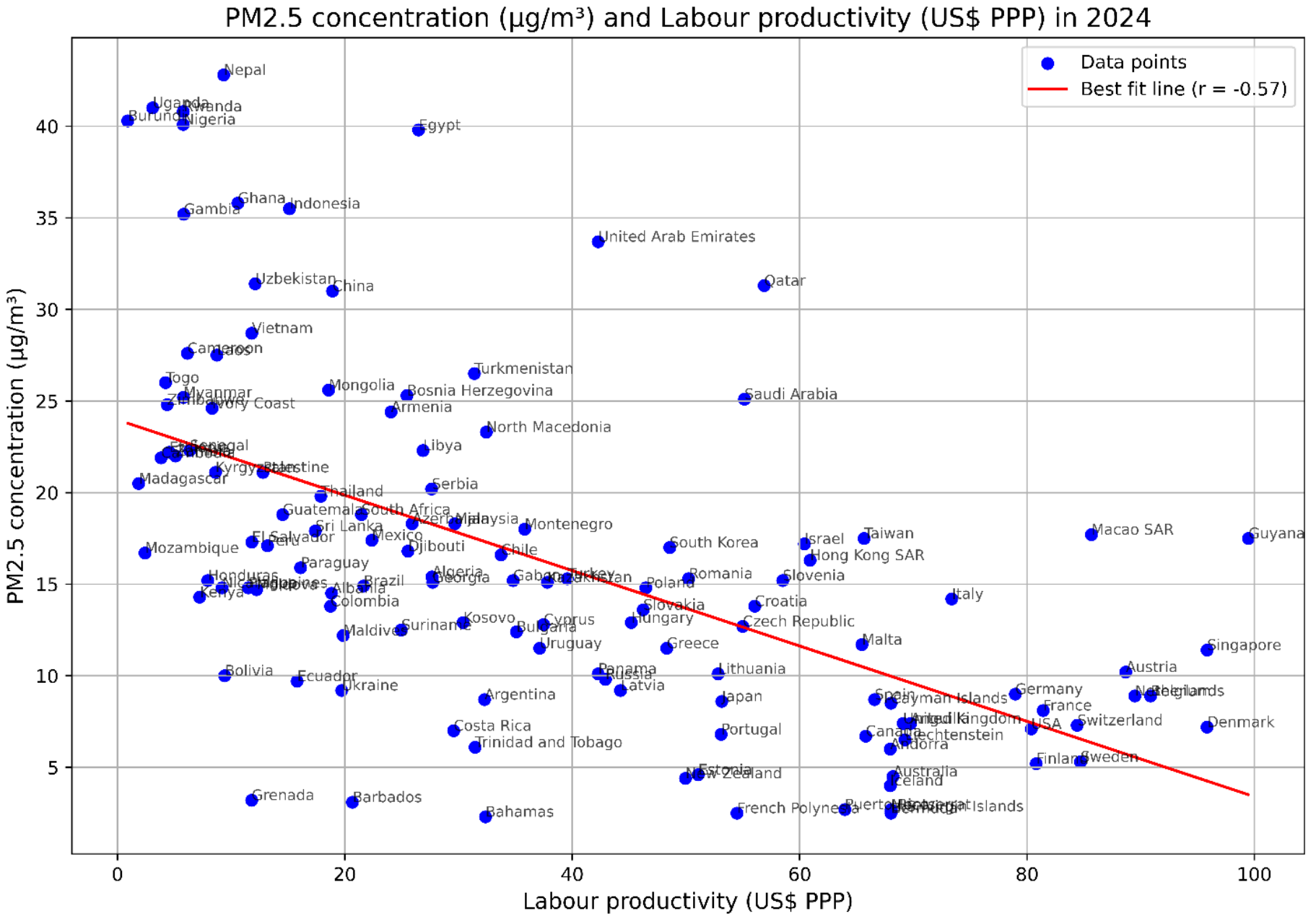
# Data

- **138 countries**
- **2022,2023,2024**
- **Pollution Outliers:** Chad, Bangladesh, Pakistan, Democratic Republic of the Congo, India, Tajikistan, Kuwait, Bahrain and Iraq
- **Productivity outliers:** Luxembourg, Norway and Ireland

| Year | 2024 | |
|---|---|---|
| Country | PM2.5 | Labour productivity |
| Bahamas | 2.3 | 32.36 |
| French Polynesia | 2.5 | 54.48 |
| Bermuda | 2.5 | 68.03 |
| U.S. Virgin Islands | 2.6 | 68.03 |
| Montserrat | 2.7 | 68.03 |
| Puerto Rico | 2.7 | 63.97 |
| Barbados | 3.1 | 20.66 |
| Grenada | 3.2 | 11.8 |
| Iceland | 4 | 67.96 |
| New Zealand | 4.4 | 49.95 |
| Australia | 4.5 | 68.2 |
| Estonia | 4.6 | 51.08 |
| Finland | 5.2 | 80.8 |
| Sweden | 5.3 | 84.7 |
| Andorra | 6 | 67.96 |
| Trinidad and Tobago | 6.1 | 31.44 |
| Liechtenstein | 6.5 | 69.26 |
| Canada | 6.7 | 65.81 |
| Portugal | 6.8 | 53.09 |
| Costa Rica | 7 | 29.57 |
| USA | 7.1 | 80.36 |
| Denmark | 7.2 | 95.83 |


PM2.5 concentration (µg/m³) and Labour productivity (US$ PPP) in 2022
Data points
Best fit line (r = -0.46)
PM2.5 concentration (µg/m³)
Labour productivity (US$ PPP)
0
20
40
60
80
100
10
20
30
40
Egypt
United Arab Emirates
Rwanda
Qatar
Saudi Arabia
Nepal
Uganda
Nigeria
Uzbekistan
Bosnia Herzegovina
Armenia
Ethiopia
Kyrgyzstan
Ghana
Indonesia
China
Mongolia
Laos
Vietnam
North Macedonia
Gabon
Zambia
Myanmar
Serbia
Madagascar
Peru
South Africa
Croatia
Kazakhstan
Chile
Turkmenistan
Turkey
Senegal
Sri Lanka
Mexico
Azerbaijan
Guatemala
Thailand
Bulgaria
Greece
South Korea
Israel
Italy
Algeria
Malaysia
Georgia
Romania
Poland
Colombia
Montenegro
Cyprus
Macao SAR
Slovenia
Philippines
El Salvador
Albania
Kosovo
Slovakia
Hong Kong SAR
Singapore
Hungary
Guyana
Malta
Brazil
Kenya
Uruguay
Russia
France
Germany
Netherlands
Maldives
Honduras
Spain
Switzerland
Latvia
Ukraine
Nicaragua
Panama
Japan
United Kingdom
USA
Denmark
Cambodia
Costa Rica
Argentina
Portugal
Liechtenstein
Bolivia
Canada
Suriname
Sweden
Andorra
Trinidad and Tobago
Finland
Puerto Rico
Australia
Grenada
Iceland
French Polynesia


PM2.5 concentration (µg/m³) and Labour productivity (US$ PPP) in 2023
Data points
Best fit line (r = -0.53)
PM2.5 concentration (µg/m³)
Labour productivity (US$ PPP)
0
20
40
60
80
5
10
15
20
25
30
35
40
Nepal
Egypt
United Arab Emirates
Qatar
Rwanda
Indonesia
China
Libya
Vietnam
Uzbekistan
Bosnia Herzegovina
Armenia
Saudi Arabia
North Macedonia
Thailand
Cambodia
Mongolia
Malaysia
Kazakhstan
Montenegro
Madagascar
Serbia
Turkey
Taiwan
South Africa
El Salvador
Sri Lanka
Azerbaijan
Chile
South Korea
Israel
Greece
Guyana
Togo
Ivory Coast
Albania
Georgia
Gabon
Romania
Hong Kong SAR
Macao SAR
Slovenia
Italy
Maldives
Honduras
Philippines
Colombia
Algeria
Cyprus
Poland
Croatia
Singapore
Bolivia
Brazil
Bulgaria
Slovakia
Kosovo
Uruguay
Hungary
Malta
Czech Republic
Kenya
Suriname
Lithuania
Canada
Spain
Russia
Panama
Japan
France
Austria
Belgium
Switzerland
Netherlands
Argentina
Ukraine
Latvia
Andorra
United Kingdom
Denmark
Ecuador
Portugal
Costa Rica
Trinidad and Tobago
Bahamas
Sweden
Finland
Estonia
New Zealand
Australia
Grenada
French Polynesia


PM2.5 concentration (µg/m³) and Labour productivity (US$ PPP) in 2024
Data points
Best fit line (r = -0.57)
PM2.5 concentration (µg/m³)
Labour productivity (US$ PPP)
0
20
40
60
80
100
5
10
15
20
25
30
35
40
Nepal
Uganda
Rwanda
Burundi
Nigeria
Egypt
Ghana
Gambia
Indonesia
United Arab Emirates
Uzbekistan
China
Qatar
Vietnam
Cameroon
Togo
Turkmenistan
Mongolia
Bosnia Herzegovina
Saudi Arabia
Myanmar
Ivory Coast
Armenia
North Macedonia
Libya
Madagascar
Serbia
Thailand
Guatemala
South Africa
Sri Lanka
Montenegro
Mexico
Djibouti
South Korea
Israel
Taiwan
Macao SAR
Guyana
Mozambique
Chile
Hong Kong SAR
Paraguay
Honduras
Algeria
Brazil
Romania
Slovenia
Kenya
Albania
Colombia
Poland
Italy
Slovakia
Croatia
Kosovo
Cyprus
Hungary
Czech Republic
Maldives
Suriname
Bulgaria
Uruguay
Greece
Malta
Singapore
Bolivia
Ecuador
Ukraine
Panama
Russia
Lithuania
Austria
Latvia
Argentina
Japan
Spain
Germany
Cayman Islands
France
Switzerland
Denmark
Costa Rica
Portugal
Trinidad and Tobago
Andorra
Finland
Sweden
Estonia
New Zealand
Australia
Iceland
Grenada
Barbados
Bahamas
French Polynesia

# Results

- P-value under 0.05 suggesting correlation between the concentration of PM2.5 and labour productivity.
- More productive countries tend to have cleaner air.

| Year | 2022 |
|---|---|
| Correlation coefficient | 0.457711479 |
| R Squared | 0.209499798 |
| Intercept | 24.46884366 |
| Slope | -0.187742976 |
| P value | 4.97192E-07 |
| Standard Error | 9.30104086 |

| Year | 2023 |
|---|---|
| Correlation coefficient | 0.531622601 |
| R Squared | 0.28262259 |
| Intercept | 24.02487391 |
| Slope | -0.191959928 |
| P value | 6.94336E-10 |
| Standard Error | 7.960423087 |

| Year | 2024 |
|---|---|
| Correlation coefficient | 0.573003939 |
| R Squared | 0.328333514 |
| Intercept | 23.96789092 |
| Slope | -0.20571059 |
| P value | 2.36012E-12 |
| Standard Error | 7.887937008 |